

AgriAI Advisor: An Intelligent Crop Disease Detection and Advisory System for Precision Agriculture in Pakistan

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Abstract—Agriculture is the backbone of Pakistan’s economy, yet farmers face significant challenges in identifying crop diseases due to lack of expertise and timely advisory services. This project presents AgriAI Advisor, an AI-powered crop disease detection and advisory system that integrates workflow automation, weather data, and large language models (LLMs) to provide real-time, localized recommendations. The system leverages n8n for orchestration, Groq API for intelligent reasoning, and OpenWeather API for environmental context. Additionally, a user-friendly web interface allows farmers to input crop problems and receive structured reports including disease identification, causes, treatment, prevention strategies, and risk analysis. The system also supports Urdu translation for accessibility. Experimental results demonstrate that the system provides fast, reliable, and context-aware agricultural guidance, making it a scalable solution for smart farming in developing regions.

I. INTRODUCTION

Agriculture contributes approximately 19

With the rise of Artificial Intelligence (AI), automated systems can assist in early detection and advisory. This project aims to bridge the gap by developing an intelligent system that provides instant crop disease diagnosis and actionable recommendations.

II. PROBLEM STATEMENT

Farmers in Pakistan lack access to:

- Real-time crop disease diagnosis
- Localized treatment recommendations
- Easy-to-use digital tools

This results in delayed decision-making and increased crop damage.

III. OBJECTIVES

- Develop an AI-based crop disease detection system
- Integrate weather data for contextual analysis
- Provide localized treatment and prevention suggestions
- Build an intuitive web-based user interface
- Automate the entire pipeline using n8n workflows
- Support Urdu language for farmer accessibility

IV. SYSTEM ARCHITECTURE

The system follows a modular pipeline architecture:

Input → Processing → AI Analysis → Localization → Output

A. Architecture Components

- **Frontend UI:** HTML, CSS, JavaScript
- **Backend Automation:** n8n Workflow Engine
- **AI Engine:** Google groq API
- **External Data:** OpenWeather API

V. METHODOLOGY

A. Input Phase

Users provide crop name and problem description via web interface.

B. Data Preprocessing

Input is normalized and structured using n8n function nodes.

C. Data Enrichment

Weather data (temperature, humidity) is fetched to improve AI predictions.

D. AI Analysis

Four AI modules are used:

- Disease Identification
- Cause Analysis
- Treatment Suggestion
- Prevention Strategies

E. Localization

Output is translated into Urdu using AI to ensure accessibility for local farmers.

F. Output Generation

Results are formatted into a structured report and displayed in UI.

VI. IMPLEMENTATION

A. Tools and Technologies

- n8n (Workflow Automation)
- Google Groq API (LLM)
- OpenWeather API
- HTML/CSS/JavaScript (Frontend)

B. Workflow Design

The workflow consists of 17 nodes including:

- Webhook Trigger Node
- Set and Function Nodes
- HTTP Request Nodes (API calls)
- AI Processing Nodes
- Merge Node
- Respond to Webhook Node

C. Algorithm

- 1) Receive user input
- 2) Normalize data
- 3) Fetch weather information
- 4) Perform AI-based disease detection
- 5) Generate cause, treatment, prevention
- 6) Calculate risk score
- 7) Translate output into Urdu
- 8) Return structured report

VII. RESULTS AND DISCUSSION

The system successfully generates detailed crop advisory reports in real-time. The output includes:

- Disease Identification
- Risk Level (60–100)
- Cause Explanation
- Treatment Recommendations
- Prevention Measures

The UI presents results in a clean and readable format, improving usability.

VIII. USER INTERFACE

The system includes a web-based interface that allows users to:

- Enter crop details
- Submit queries
- View structured AI-generated reports

The UI is designed with simplicity and accessibility in mind.

IX. ADVANTAGES

- Real-time decision support
- Cost-effective solution
- Easy to use
- Localized recommendations
- Scalable architecture

X. LIMITATIONS

- Requires internet connectivity
- AI responses may vary
- Limited dataset for certain crops

XI. FUTURE WORK

- Image-based disease detection (Computer Vision)
- Mobile application development
- Integration with WhatsApp/Telegram bots
- Database for historical analysis
- IoT sensor integration for real-time monitoring

XII. CONCLUSION

This project demonstrates the effective use of Artificial Intelligence and workflow automation to address real-world agricultural challenges. The AgriAI Advisor provides a scalable and intelligent solution for crop disease detection and advisory, contributing towards smart farming initiatives in Pakistan.

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REFERENCES

- [1] OpenWeather API Documentation. Available: <https://openweathermap.org>
- [2] Google Groq API Documentation
- [3] n8n Workflow Automation Platform